# RAG PLAYBOOK REPORT

The learning lab playbook showed us why RAG is better than LLM's for getting updated information. I've built a RAG system by following the RAG Pipeline step by step. I learnt that the documents loaded should be related to get better RAG answers. The chunk sizes matter because context can be lost in case of small chunks and in case of large chunks, unnecessary information is given. Then the chunks are embedded (turned into vectors) and similar chunks are embedded together.

These embeddings are stored into vectors and are indexed for easy retrieval, in vector database related words are searched instead of exact match. Popular vector DBs: FAISS (local, fast), ChromaDB (easy, small scale), Pinecone (cloud), Weaviate (production). And when user asks a query, related chunks are retrieved and a prompt is added and forwarded to the LLM for final output.

From Advanced Playbook, I've learned that RAG Architecture is a two-phase architecture, Offline Indexing- Local tasks, Online Retrieval- Query retrieval, related chunks retrieval, LLM prompt. I've also learned how temperature settings determine the hallucinations given by LLM's. By sequencing the order of RAG Pipeline, I clearly understood the workflow. I also learned that poor chunking results in bad retrieval of related chunks.

I've also seen the comparison of Chunking strategies – fixed size, recursive character splitting, semantic chunking, document-specific. I also learned about overlapping and that it helps in preserving the context when multiple chunks are retrieved. I learned about different Embedding Models like all-MiniLM-L6-v2, text-embedding-ada-002 (OpenAI) etc and how they are used based on requirements. I now understand that high cosine score means more similarity and only chunks with high cosine score are retrieved.

This playbook also specifies the retrieval failures that can happen - like wrong chunks being retrieved, answer not being present in the retrieved chunks, LLM ignoring the chunks, hallucinations of LLM, high latency. I also learned how RAG Systems are evaluated using the metrics like faithfulness, answer relevancy, context precision, context recall. All the above evaluation metrics should have high score for good performance of the RAG system.

Overall, both the playbooks helped me grasp the concept of RAG Pipeline and now I'm confident in building a RAG System.